BDP-insight

L2 HodgeConverter

API name: BDP-insight-L2 · whitebox MRI

v0.1.1 Public Prototype — English Edition

User & Experimenter Guide

API Name	BDP-insight-L2
Track	Meta-13 Debugger / BDP-insight prototype
Test Models	Qwen/Qwen2.5-0.5B-Instruct (baseline) + Gemma4-4ae (~7B effective, compatibility)
Prepared	2026-05-06
Source	Internal Dynamic Debugger experiment logs

This document is the English-language user/experimenter guide for the v0.1.1 public prototype. Some capabilities (direct L2+L3 representation/tensor surgery, residual graft, deep hidden-state intervention) are intentionally omitted to reduce misuse risk. A Korean-language reference PDF is shipped as an appendix in the package (kr_appendix/).

⚠️ IMPORTANT NOTICE ⚠️

Behavior varies BY MODEL and BY INPUT LANGUAGE.

Even with identical prompt and regex settings, prefix-exposure patterns differ across models and across input languages (e.g. English vs Korean).

Do not rely on the figures in this document — please reproduce the experiments in your own environment before drawing conclusions.

■ API Hub Listing — BDP-insight-L2

BDP-insight provides a prototype HodgeConverter API (BDP-insight-L2) for controlled LLM runtime-intervention experiments.
The API routes selected token / value / symbol-level intervention requests through an externalized server kernel and returns a compact public result shape.

It is designed for:

- HodgeConverter tutorial reproduction
- symbolic redirection experiments
- runtime intervention testing
- BDP-insight / Meta-13 debugger prototype analysis

This API is not intended for bulk text processing. Claims are prototype-level and observational. It does not guarantee full control over model reasoning.
Website will be open soon.

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1. Product description

BDP-insight-L2 HodgeConverter exposes a controlled output-prefix intervention path. The original prompt text is not rewritten. The debugger observes partial generated output, sends the selected target span to the server-bound Hodge kernel, and resumes continuation from the patched prefix.

This public prototype is neither a guaranteed arithmetic-correction system nor a general reasoning booster. It must be presented as a controlled research and debugging tool.

2. How it works (Output-Prefix Intervention)

2.1 Dynamic L2 flow

```
user prompt
-> model begins generation
-> partial assistant output appears
-> target_first_breakpoint checks target_regex
-> server-bound Hodge kernel patches the target span
-> local debugger resumes from the patched prefix
-> final continuation reflects the modified prefix,
   base model ability, and prompt pressure
```

2.2 Decisive difference from prompt rewriting

```
prompt text unchanged   = true
output prefix mutated   = true
continuation resumes from patched prefix
```

The input prompt is never altered. Only one fragment of the model's own partial output is modified. Continuation then resumes from that mutated prefix.

3. Release boundary, privacy & API credentials

Tier	Includes	Notes
Free / Public	Tutorial reproduction, regex recipes, basic L2 dynamic-debugger flow	Limited rate; intended for evaluation
Paid (server-bound)	Full Hodge kernel via API Hub; richer regex/wait/cont profiles	Routes through externalized server kernel; compact public result shape
Future research only	Direct L2+L3 representation/tensor surgery, residual graft, deep hidden-state intervention	Intentionally OMITTED from v0.1.1 to reduce misuse risk

3.1 Privacy notice

- Prompt/token observations or partial generated output may be transmitted to the paid server kernel.
- Do not submit secrets, credentials, personal data, private datasets, or confidential production prompts.
- BDP does not intentionally persist raw prompts or generated text; request payloads are discarded after the operation.
- Operational logs are limited to request IDs, timestamps, status, usage counters, lengths, hashes, and plan metadata.

3.2 API credentials — customer vs provider

Two credential surfaces exist. Keep them strictly separated. The customer-side credentials are what end users send with every request. The provider-side proxy secret is held by BDP only and must never be displayed in customer-facing documentation, examples, or client SDKs.

Side	Header	Who holds it	Visibility
Customer	X-RapidAPI-Key	End user (subscriber)	Required in every request from the customer
Customer	X-RapidAPI-Host	End user (subscriber)	Required in every request from the customer
Provider	X-RapidAPI-Proxy-Secret	BDP (provider) only	MUST NOT appear in customer docs, examples, or SDKs

Treat any leakage of X-RapidAPI-Proxy-Secret as a credential incident: rotate the secret on the provider side immediately and audit recent traffic.

4. Quick Start — first experiment in 5 minutes

#	What to do	Example	Why
1	Send a prompt that exposes an intermediate formula	"What is 2 + 4?" / "First write the equation, then compute."	The target must appear in partial output before it can be patched
2	auto new_word	9 (replaces left operand)	The value that overwrites the matched span
3	target_regex	(?<=\+)\s*\d+	Regex pinpointing the patch location
4	wait_tokens / continuation_tokens	16 / 32 (arithmetic default)	wait = tokens until the target appears; continuation = tokens generated after the patch
5	Run	2 + 4 = 6 → 2 + 9 = 11	target-first breakpoint reduces stale-tail artifacts
6	5-stage decision	(1) target detection (2) server patch (3) prefix mutation (4) continuation effect (5) final answer	Do not evaluate by the final answer alone

4.1 Formula-guiding wording (recommended)

```
First write only the equation, then compute it.

Write the equation first, then calculate.

Complete only the equation: ...

You MUST first write a single-line equation of the form "5 + 2 =", then continue
with the calculation.
```

4.2 Common pitfalls

- **target not exposed:** If the model emits only the answer, the target never appears in the partial output. Use formula-guided wording.
- **evaluating only final answer:** A weak local model may fail arithmetic even without intervention. Apply the 5-stage decomposition.
- **template pressure:** Some natural-language templates (in any language) may reorder operands after the patch (see EXP-A02 / A03).
- **breakpoint too early:** If wait_tokens is too small, output may be cut mid-phrase (e.g. 'The sum of 2'), breaking meaning.
- **result-token stale tail:** Without a target-first breakpoint, the old answer may persist.
- **pre-existing distribution clash:** Setting new_word to a value already common in the distribution can cause order swaps (EXP-A02, hypothesis 1).

5. Experiment summary (29 records: Qwen 25 + Gemma 4)

5.1 Status counts — Qwen/Qwen2.5-0.5B-Instruct

Status	Count
Total logged experiments	25
Success / observed influence	18
No target / inactive	3
Base / model limitation	2
Breakpoint / tail artifact	1
Needs review	1

5.2 Category split

Category	Records	Source model
Arithmetic	11	Qwen 0.5B
Mid-value substitution	5	Qwen 0.5B
Word-problem math	4	Qwen 0.5B
Reasoning	5	Qwen 0.5B
Cross-model compat	4	Gemma4-4ae (~7B effective)

5.3 Cross-model compatibility — Gemma4-4ae

Four additional records (EXP-GEMMA-01..04) confirm that the dynamic debugger flow is not Qwen-specific. Detailed per-record notes appear in §11 and Appendix A.

6. Key success cases

EXP-A05 · Arithmetic / left operand swap

Prompt: 'Compute 2 + 4'. Patch left operand 2 → 9. Continuation: 2 + 9 = 11. Demonstrates the canonical L2 path on the simplest arithmetic surface.

EXP-A06 · Arithmetic / right operand swap

Patch right operand. Same flow, mirrored. Validates symmetry of the regex/new_word pair.

EXP-A07 · Arithmetic / operator swap

Patch '+' → '-'. Result mutates predictably; prompt text unchanged.

EXP-A08 · Arithmetic / two-step pressure

Formula-guided two-step prompt. Patch on first operand; second-step computation tracks.

EXP-A09 · Arithmetic / multi-digit

Multi-digit operand. Confirms the regex generalizes to longer numeric spans.

EXP-B02 · Mid-value substitution

Patch an intermediate value inside a chain-of-arithmetic. Continuation re-derives downstream values from the mutated prefix.

7. Observed behavior classes

7.1 Within-model classes (Qwen 0.5B baseline)

#	Class	Description
6.1	Direct surface success	Target detected, patched, continuation reflects the patched prefix.
6.2	Template pressure swap	Continuation re-orders operands under natural-language template pressure.
6.3	Stale tail	Old answer persists due to absent target-first breakpoint.
6.4	Base limitation	Local model fails the task even without intervention.
6.5	No-target run	Target never appears in the partial output.
6.6	Distribution clash	new_word equals an already-common token; ordering swaps unexpectedly.

7.2 Cross-model classes (observational)

#	Class	Description
7.1	Cross-model compatibility success	The dynamic-debugger flow operates on a different model family without code changes.
7.2	Structured-output intervention surface	Models that emit structured intermediate scaffolding (problem analysis / setup / equation / computation / answer) expose a wider intervention surface.
7.3	Formula-prefix structural recomposition	Patching a formula-level token can cause the model to recompose the surrounding equation (e.g. operand swap with operator change).
7.4	Scaffold-aided stable detection	Auto-generated step scaffolds make target detection more stable in some configurations.

8. Caveats and limitations

- **Avoid claim — guaranteed reasoning correction:** This system is not a guaranteed reasoning corrector.
- **Avoid claim — universal arithmetic improvement:** No claim is made of universal arithmetic improvement.
- **Avoid claim — full hidden-state control:** L2 operates on the token surface, not on hidden state directly.
- **Avoid claim — proof of tensor causality:** No causal proof of tensor-level intervention is asserted.
- **Single-model evidence:** Most evidence is on a small Qwen baseline; cross-model evidence is observational.
- **Language variability:** Behavior may vary substantially across input languages.
- **Stale tail / breakpoint sensitivity:** Outcomes are sensitive to wait_tokens, continuation_tokens, and breakpoint placement.

9. Regex recipes

Goal	target_regex	new_word	wait / cont
Replace the right operand of a + b	(?<=\+)\s*\d+	9	16 / 32
Replace the left operand of a + b	\d+(?=\s*\+)	9	16 / 32
Replace the operator + → -	\+	-	16 / 32
Replace a multi-digit operand	\d{2,}	99	16 / 32
Patch an intermediate result (= n)	(?<==\s)\d+	7	16 / 32
Match an equation prefix 'a + b ='	\d+\s*\+\s*\d+\s*=	5 + 2 =	32 / 32
Patch a step label inside a scaffold	(?<=Step\s)\d	3	32 / 32

10. Static debugger / IRSDCE follow-up note

These dynamic-debugger experiments should later be paired with the static debugger's IRSDCE views. The static view can illustrate how output-prefix intervention shifts hidden-state geometry, layer-wise flow, and token-conditioned trajectories. It must be documented as an observability layer, not as a causal proof in itself.

```
Static MRI / IRSDCE View:
Layer-wise Hidden-State Geometry
Under Token-Prefix Intervention
```

11. Cross-model compatibility — Token-Surface Conversion Hypothesis

11.1 Compatibility note (observational)

The prototype is not Qwen-specific. Qwen is used as the baseline public test model, while Gemma-style structured outputs show that models exposing more intermediate scaffolding can provide a wider intervention surface. This is an observational compatibility note, not a guarantee.

11.2 Token-Surface Conversion Hypothesis (H1)

```
H1 (HYPOTHESIS, not a theorem):
Some models perform a larger share of their reasoning by writing
intermediate steps onto the token surface, rather than completing
them implicitly inside hidden state.

[Mechanism]
Such models tend to emit explicit scaffolds:
problem analysis -> setup -> equation -> computation -> answer.

[Relation to Hodge L2]
L2 HodgeConverter operates on the token surface. As a result,
the more a model converts reasoning into surface tokens,
the wider the surface available for L2 intervention.
```

Note: this is NOT a claim that 'larger models produce more tokens'. The hypothesis concerns the SHARE of reasoning that surfaces as tokens versus the share that completes implicitly inside hidden state. When the surface share is higher, the operative surface for L2 widens — regardless of parameter count.

11.3 Gemma4-4ae (~7B effective) compatibility records — 4 cases

ID	Sub-category	Change	Before	After	Judgment
EXP-GEMMA-01	structured output / box problem	2 → 9	boxes: 2 / formula 2 × 3 / answer 6	boxes: 9 (if there were 9 boxes...)	Cross-model compat + structured-output surface
EXP-GEMMA-02	structured output / continued reasoning	2 → 9	(same as GEMMA-01)	## Problem Analysis / 9 boxes each containing 3 apples...	Cross-model compat + formula-guided continuation
EXP-GEMMA-03	formula-prefix structural recomposition	2 → 9	equation $2 + 3$ / compute 5	equation $9 - 2$ / compute 7	Formula-prefix structural recomposition

EXP-GEMMA-04	natural-language scaffold	2 → 9	Cheolsu 2 / received 3 / answer 5	Cheolsu initially had 9 / received 3 / ...	Cross-model compat + structured-output surface

11.4 Qwen 0.5B vs Gemma4-4ae — observational comparison

Axis	Qwen 0.5B	Gemma4-4ae (~7B effective)
Intermediate output surface	Short, compressed; often emits answer only.	Explicit step-wise scaffolding.
Target detection stability	Wobbly under mixed natural-language/numeric (EXP-D01).	Stable, scaffold-aided (EXP-GEMMA-04).
Intervention surface	Narrow; formula-guided wording usually required.	Wider; already exposed in base output.
Base arithmetic limit	Weak (EXP-A11 fails arithmetic itself).	Stable; under prefix pressure may recompose (EXP-GEMMA-03).
Notable case	Token-order swap (EXP-A02).	Formula-prefix structural recomposition (EXP-GEMMA-03).
L2 operating mode	Minimal validation only.	Provides supporting evidence for the H1 hypothesis.

11.5 One-sentence conclusion

```
The Gemma records support the view that HodgeConverter is
not a Qwen-specific trick but a dynamic-debugger structure
applicable across models whose mid-generation prefix is observable;
and where more reasoning surfaces as scaffolded tokens, the
operative surface for L2 intervention naturally widens.
This remains an observational hypothesis, not a guarantee.
```

12. Glossary

Term	Definition
Output-prefix intervention	Modifying a fragment of the model's own partial output, then resuming generation from the modified prefix; the input prompt is unchanged.
target_regex	Regular expression that pinpoints which span of the partial output to patch.
new_word	The replacement value applied to the matched span.
wait_tokens	Tokens to wait while watching for the target to appear in the partial output.
continuation_tokens	Tokens generated after the patch.
target_first_breakpoint	Strategy of pausing as soon as the target appears, before the model emits a stale tail.
Hodge kernel (server-bound)	Externalized server-side component that performs the patch and returns a compact public result shape.
Compact public result shape	The minimized, redacted return payload visible to the customer.
L2	Token-surface intervention layer (this prototype).
L2+L3 (omitted)	Direct representation/tensor surgery, residual graft, deep hidden-state intervention; intentionally not exposed in v0.1.1.
Stale tail	Old result text persisting after a patch when no target-first breakpoint is used.
Distribution clash	new_word coincides with an already-common token, causing unexpected re-orderings.
Token-Surface Conversion (H1)	HYPOTHESIS that some models convert more reasoning into surface tokens, widening the L2 intervention surface.
Intervention surface	The portion of model output observable enough to be patched.
Cross-model compat	Observational evidence that the L2 path runs on more than one model family.
Structured-output surface	Output style that exposes problem analysis / setup / equation / computation / answer.
Formula-prefix structural recomposition	Phenomenon where patching a formula-level token causes the surrounding equation to be recomposed.
Scaffold-aided detection	Detection that becomes more stable thanks to auto-generated step labels.
IRSDCE	Static debugger view family for layer-wise hidden-state geometry under token-prefix intervention (observability only).
X-RapidAPI-Proxy-Secret	Provider-side proxy secret. Held by BDP only; MUST NOT be shown in customer-facing material.

Appendix A — Full experiment table (29 rows)

Note: Some rows preserve original Korean prompt/output text verbatim because the experiments were run with Korean prompts. This is intentional — preserving raw experiment data is more honest than retroactive translation.

ID	Category	Change	target_regex	wait/cont	Before	After	Status
EXP-A01	Arithmetic / no-target run	—	(?<=\+)\s*\d+	16 / 32	answer only	no target	no target / inactive
EXP-A02	Arithmetic / token-order swap	2 -> 5	\d+(?=\s*\+)	16 / 32	2 + 3 = 5	5 + 2 = 7 (operand swap)	needs review
EXP-A03	Arithmetic / template pressure	2 -> 9	\d+(?=\s*\+)	16 / 32	The sum of 2 and 3 is 5	The sum of 9 and 3 is 12 (order shifts)	success / observed influence
EXP-A04	Arithmetic / breakpoint too early	2 -> 9	(?<=\+)\s*\d+	8 / 32	The sum of 2	(cut mid-phrase)	breakpoint / tail artifact
EXP-A05	Arithmetic / left operand swap	2 -> 9	\d+(?=\s*\+)	16 / 32	2 + 4 = 6	2 + 9 = 11	success / observed influence
EXP-A06	Arithmetic / right operand swap	4 -> 9	(?<=\+)\s*\d+	16 / 32	2 + 4 = 6	2 + 9 = 11	success / observed influence
EXP-A07	Arithmetic / operator swap	+ -> -	\+	16 / 32	2 + 4 = 6	2 - 4 = -2	success / observed influence
EXP-A08	Arithmetic / two-step pressure	2 -> 9	\d+(?=\s*\+)	32 / 32	2 + 3 = 5; 5 + 4 = 9	9 + 3 = 12; 12 + 4 = 16	success / observed influence
EXP-A09	Arithmetic / multi-digit	12 -> 99	\d{2,}	16 / 32	12 + 7 = 19	99 + 7 = 106	success / observed influence
EXP-A10	Arithmetic / distribution clash	5 -> 5	(?<=\+)\s*\d+	16 / 32	2 + 5 = 7	5 + 2 = 7 (re-ordered)	success / observed influence
EXP-A11	Arithmetic / base failure	—	(?<=\+)\s*\d+	16 / 32	model arithmetic wrong	still wrong	base / model limitation
EXP-B01	Mid-value / chain swap	5 -> 9	(?<==\s)\d+	16 / 32	= 5; next 5+1 = 6	= 9; next 9+1 = 10	success / observed influence
EXP-B02	Mid-value / explicit step	5 -> 9	(?<==\s)\d+	16 / 32	step1 = 5	step1 = 9 (downstream tracks)	success / observed influence
EXP-B03	Mid-value / negative	5 -> -3	(?<==\s)-?\d+	16 / 32	= 5	= -3	success / observed

							influence
EXP-B04	Mid-value / float	5 -> 5.5	(?<==\s)\d+	16 / 32	= 5	= 5.5 (continuation tracks)	success / observed influence
EXP-B03-R1	Mid-value / negative (replay)	5 -> -3	(?<==\s)-?\d+	32 / 32	= 5	= -3 (stable replay)	success / observed influence
EXP-C01	Word-problem / count change	2 -> 9	\d+	16 / 32	boxes 2 -> answer 6	boxes 9 -> answer 27	success / observed influence
EXP-C02	Word-problem / unit change	kg -> g	\bkg\b	16 / 32	10 kg	10 g (downstream calc tracks unit)	success / observed influence
EXP-C03	Word-problem / stale tail	2 -> 9	\d+	8 / 32	answer 6 already emitted	patch ignored	breakpoint / tail artifact
EXP-C04	Word-problem / no target	—	\d+	16 / 32	answer only	no target	no target / inactive
EXP-D01	Reasoning / mixed NL + number	2 -> 9	\d+	16 / 32	wobbly detection	partial influence	needs review
EXP-D02	Reasoning / scaffold prompt	step 2 -> 3	(?<=Step\s)\d	32 / 32	Step 2 ...	Step 3 ... (scaffold tracks)	success / observed influence
EXP-D03	Reasoning / answer-only prompt	—	\d+	16 / 32	no formula in output	no target	no target / inactive
EXP-D04	Reasoning / two-condition	yes -> no	\byes\b	16 / 32	yes branch	no branch	success / observed influence
EXP-D05	Reasoning / chained patch	5 -> 9	(?<==\s)\d+	32 / 32	intermediate 5	intermediate 9, downstream tracks	success / observed influence
EXP-GEMMA-01	Cross-model compat / structured output / box problem	2 -> 9	\d+	32 / 32	boxes: 2 / formula 2 × 3 / answer 6	boxes: 9 (if there were 9 boxes...)	cross-model compat
EXP-GEMMA-02	Cross-model compat / structured output / continued	2 -> 9	\d+	64 / 32	(same as GEMMA-01)	## Problem Analysis / 9 boxes each containing 3...	cross-model compat
EXP-GEMMA-03	Cross-model compat / formula-prefix recompositio	2 -> 9	\d+	64 / 32	$2 + 3$ -> 5	$9 - 2$ -> 7 (recomposed)	cross-model compat

	n						
EXP-GEMMA-04	Cross-model compat / NL scaffold	2 -> 9	\d+	32 / 32	Cheolsu 2 / received 3 / answer 5	Cheolsu initially 9 / received 3 / ...	cross-model compat

Appendix B — Changelog

- **Initial release:** v0.1.1 — initial public prototype packaging. 25 Qwen experiments + base manual + 10-sheet workbook.
- **Cross-model update:** Cross-model compatibility records added (Gemma4-4ae, ~7B effective): EXP-GEMMA-01..04. New 'Cross_Model_Compat' workbook sheet. Token-Surface Conversion stated as a hypothesis. Behavior class additions 7.1–7.4.
- **English Edition:** English Edition for API Hub / GitHub distribution. English-only README, User Guide, Workbook. R-labels removed from public title. Customer/provider credential separation added (X-RapidAPI-Proxy-Secret is provider-only). 'larger model' phrasing softened to 'more structured intermediate output → wider intervention surface'. Korean reference PDF kept under kr_appendix/.

This document is the user/experimenter guide for BDP-insight-L2 (L2 HodgeConverter), v0.1.1 Public Prototype — English Edition. For questions and feedback, please use the BDP-insight project channel.